

SET-01(Beginner)

Problem No: 01

Problem Name: Print "Hello, World!"

Source Code:

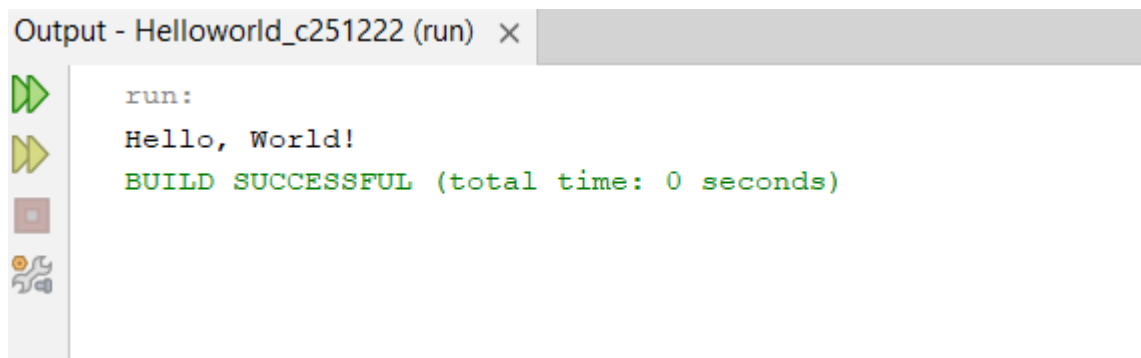
```
package helloworld_c251222;

public class Helloworld_c251222 {

    public static void main(String[] args) {

        System.out.println("Hello, World!");
    }
}
```

Output:



```
Output - Helloworld_c251222 (run) ×
run:
Hello, World!
BUILD SUCCESSFUL (total time: 0 seconds)
```

Approach:

This program represents the most basic form of a Java application and is used to understand the fundamental structure of a Java program. It does not take any input from the user. The execution starts from the main() method, which serves as the entry point of every Java program. Inside this method, the System.out.println() statement is used to print the message "Hello, World!" on the console. This statement sends the output to the standard output stream and displays it on the screen, helping beginners learn how Java programs produce output.

Problem No: 02

Problem Name: Take the user's name and greet them. Example, Input: "Your Name" and Output: Hello, "Your Name"

Source Code:

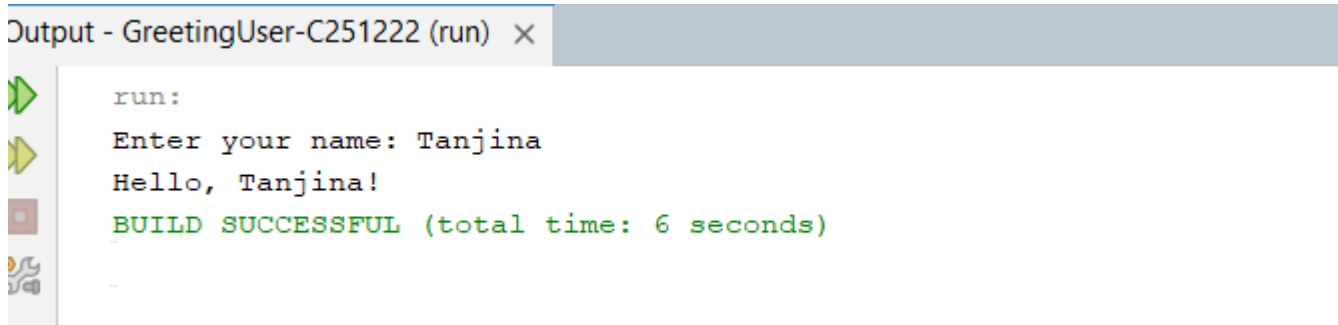
```
package greetinguser.c251222;

import java.util.Scanner;

public class GreetingUserC251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = sc.nextLine();
        System.out.println("Hello, " + name + "!");
    }
}
```

Output:



```
run:
Enter your name: Tanjina
Hello, Tanjina!
BUILD SUCCESSFUL (total time: 6 seconds)
```

Approach:

This program is a simple Java application designed to take input from the user and display a personalized greeting. It helps in understanding how user input works in Java. The execution begins from the main() method, which is the entry point of the program. Inside this method, the Scanner class is used to read input from the keyboard. The program prompts the user to enter their name using System.out.print(), and then reads the input using nextLine() method. After receiving the input, the System.out.println() statement is used to display a greeting message by combining the entered name with the text "Hello". This program demonstrates how to interact with users and produce dynamic output based on input.

Problem No: 03

Problem Name: Check if a number is even or odd

Source Code:

```
package even_odd.c251222;

import java.util.Scanner;

public class Even_oddC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int num = sc.nextInt();

        if (num % 2 == 0)

            System.out.println("Even");

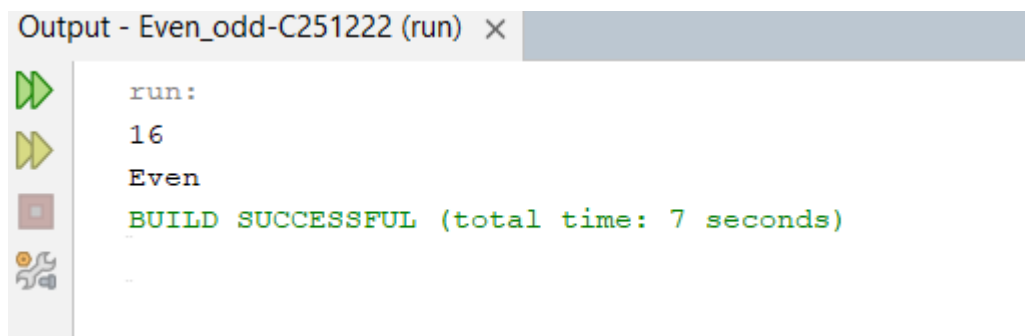
        else

            System.out.println("Odd");

    }

}
```

Output:



Approach:

This Java program is designed to determine whether a given number is even or odd. It helps in understanding user input and the use of conditional statements. The execution starts from the main() method, which is the entry point of the program. Inside this method, the Scanner class is used to take an integer input from the user. After taking the input, the program uses an if-else statement to check the condition. If the number is divisible by 2 (i.e., $\text{num} \% 2 == 0$), it prints "Even"; otherwise, it prints "Odd". The System.out.println() statement is used to display the result on the console. This program demonstrates how to use conditions to make decisions in Java.

Problem No: 04

Problem Name: Find the maximum of 3 numbers: Take 3 numbers from user input, print the largest

Source Code:

```
package maximum.c251222;

import java.util.Scanner;

public class MaximumC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();

        int max = a;

        if (b > max) max = b;

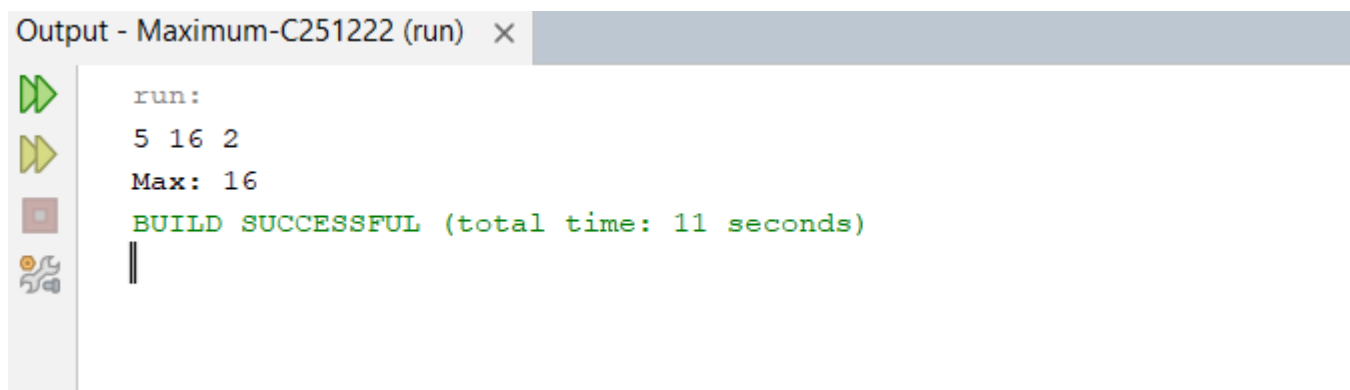
        if (c > max) max = c;

        System.out.println("Max: " + max);

    }

}
```

Output:



```
Output - Maximum-C251222 (run) ×
run:
5 16 2
Max: 16
BUILD SUCCESSFUL (total time: 11 seconds)
```

Approach:

The purpose of this program is to determine the largest number among three user-provided inputs. First, the program takes three integer values from the user using the Scanner class. It initially considers the first number as the maximum value. Then, it compares the second and third numbers with the current maximum using if conditions. If either of the numbers is greater than the current maximum, the maximum value is updated accordingly. Finally, the program prints the largest number as the output using System.out.println().

Problem No: 05

Problem Name: Simple Calculator (Add, Subtract, Multiply, Divide): Use if-else or switch to perform operations.

Source Code:

```
package calculator.c251222;

import java.util.Scanner;

public class CalculatorC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter two numbers:");

        int num1 = sc.nextInt();

        int num2 = sc.nextInt();

        System.out.println("Enter operator (+, -, *, /):");

        char op = sc.next().charAt(0);

        int result = 0;

        if (op == '+') result = num1 + num2;

        else if (op == '-') result = num1 - num2;

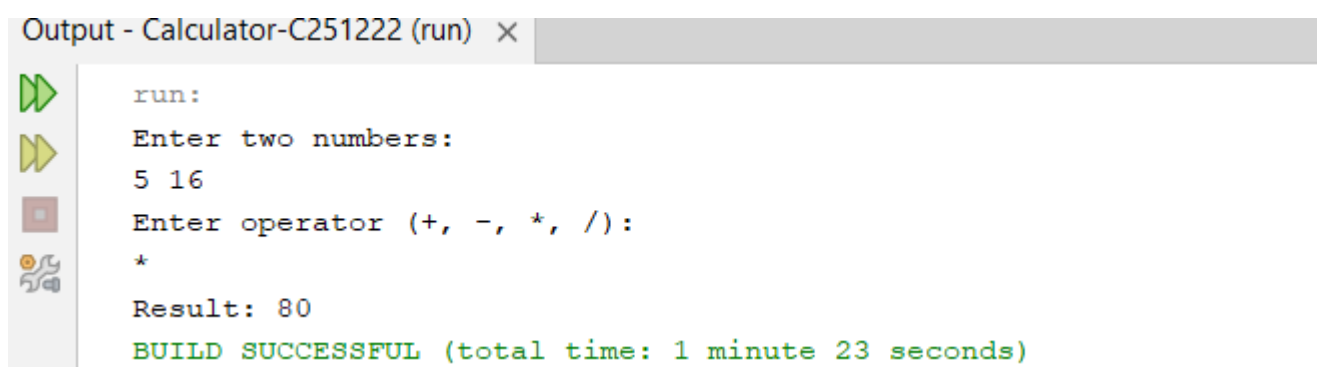
        else if (op == '*') result = num1 * num2;

        else if (op == '/') result = num1 / num2;

        System.out.println("Result: " + result);

    }
}
```

Output:



```
Output - Calculator-C251222 (run) ×
run:
Enter two numbers:
5 16
Enter operator (+, -, *, /):
*
Result: 80
BUILD SUCCESSFUL (total time: 1 minute 23 seconds)
```

Approach:

This program is a simple calculator that performs basic arithmetic operations. It takes two numbers from the user using the Scanner class and then asks for an operator (+, -, *, /). Based on the selected operator, the program uses if-else statements to perform the corresponding calculation and stores the result. Finally, it prints the result on the console using System.out.println().

Problem No: 06

Problem Name: Print numbers from 1 to N. Take N from the user input.

Source Code:

```
package print.c251222;

import java.util.Scanner;

public class PrintC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

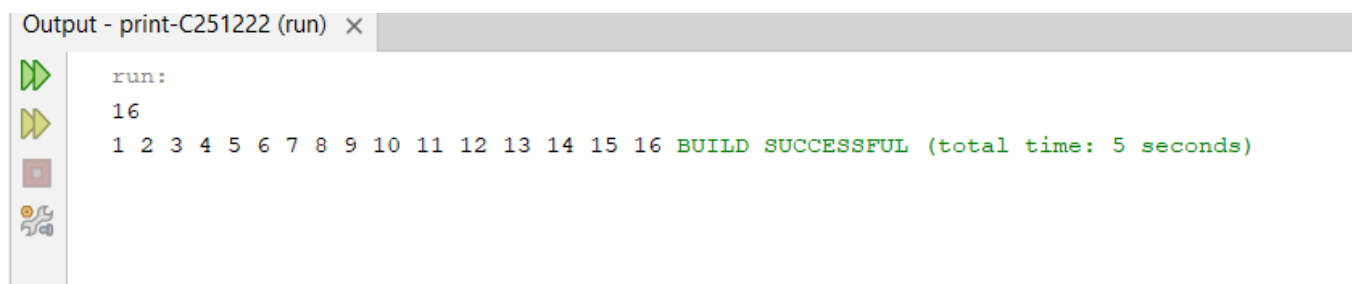
        for (int i = 1; i <= n; i++)

            System.out.print(i + " ");

    }

}
```

Output:



Approach:

This program is designed to print a sequence of numbers starting from 1 up to a user-defined value. First, it takes an integer input *n* from the user using the Scanner class. Then, a for loop is used to iterate from 1 to *n*. Inside the loop, each number is printed using `System.out.print()`, followed by a space so that all numbers appear in a single line.

Finally, the program displays all numbers from 1 to *n* as output in the console, helping to understand the use of loops and user input in Java.

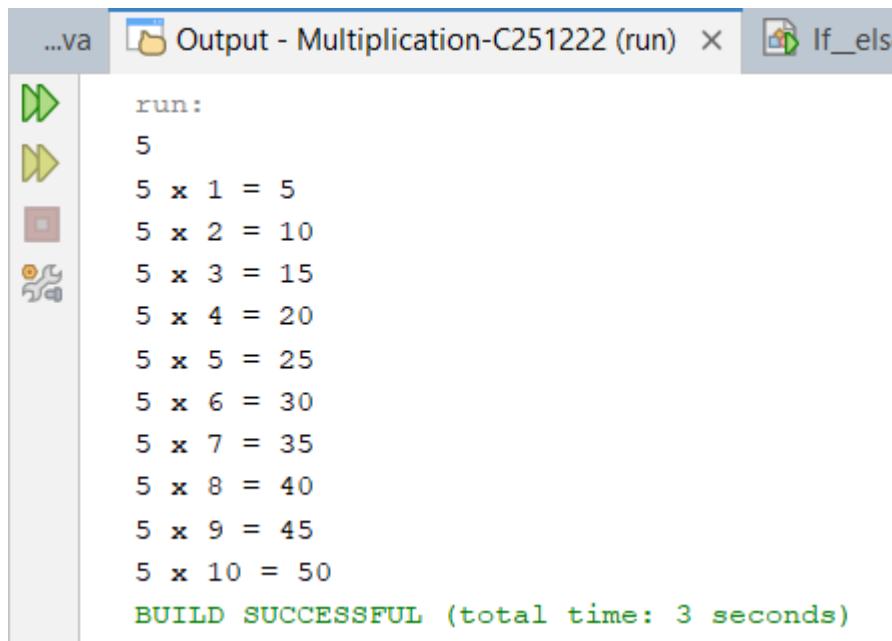
Problem No: 07

Problem Name: Print the multiplication table of a number.

Source Code:

```
package multiplication.c251222;  
  
import java.util.Scanner;  
  
public class MultiplicationC251222 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
  
        for (int i = 1; i <= 10; i++)  
            System.out.println(n + " x " + i + " = " + (n * i));  
    }  
}
```

Output:



```
run:  
5  
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50  
BUILD SUCCESSFUL (total time: 3 seconds)
```

Approach:

This program is created to generate the multiplication table of a user-provided number. First, it reads an integer value *n* from the user using the Scanner class. Then, a for loop runs from 1 to 10. Inside the loop, the program multiplies *n* with the loop variable *i* and prints the result in a formatted way using `System.out.println()`.

Finally, the program displays the multiplication table of the given number from 1 to 10 in the console, helping to understand the use of loops and arithmetic operations in Java.

Problem No: 08

Problem Name: Calculate the factorial of a number. Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Source Code:

```
package factorial.c251222;

import java.util.Scanner;

public class FactorialC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int fact = 1;

        for (int i = 1; i <= n; i++)

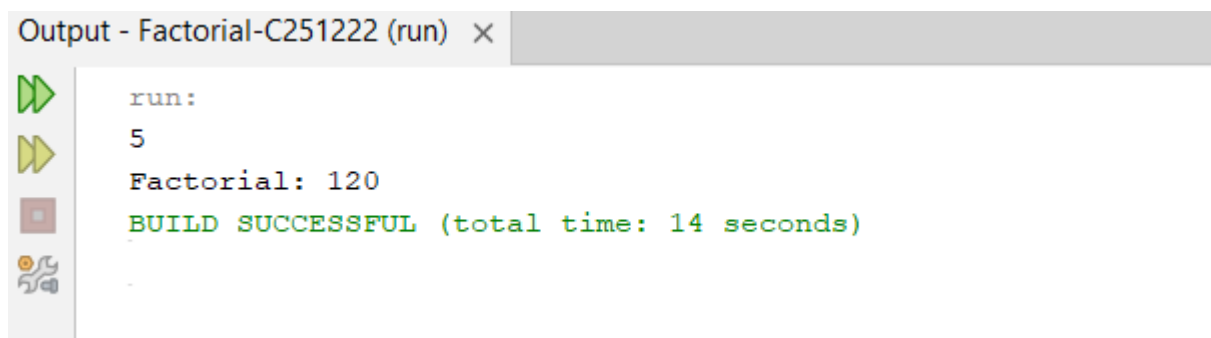
            fact *= i;

        System.out.println("Factorial: " + fact);

    }

}
```

Output:

The screenshot shows an IDE output window titled "Output - Factorial-C251222 (run)". The output text is as follows:

```
run:
5
Factorial: 120
BUILD SUCCESSFUL (total time: 14 seconds)
```

Approach:

In this program, a factorial of a number is computed. First, it takes an integer input n from the user using the Scanner class. Then, it initializes a variable $fact$ with the value 1 to store the result of the factorial calculation.

After that, a for loop is used to multiply all numbers from 1 to n . In each iteration, the value of $fact$ is updated by multiplying it with the loop variable i . Finally, the program prints the calculated factorial using `System.out.println()`, displaying the result in the console.

Problem No: 09

Problem Name: Sum of all numbers from 1 to N.

Source Code:

```
package sum;

import java.util.Scanner;

public class Sum {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= n; i++)

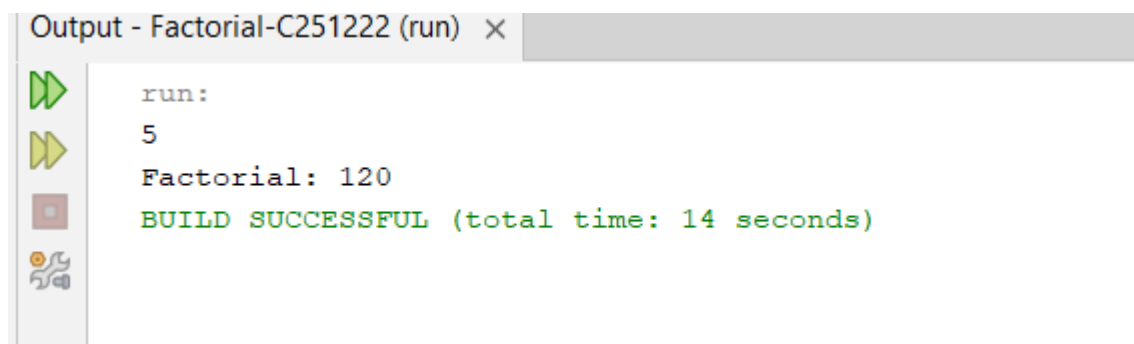
            sum += i;

        System.out.println("Sum: " + sum);

    }

}
```

Output:



Approach:

In this program, the sum of numbers from 1 to a given number is calculated. First, it takes an integer input *n* from the user using the Scanner class. Then, a variable *sum* is initialized with the value 0 to store the result of the summation.

After that, a for loop is used to add all numbers from 1 to *n*. In each iteration, the value of *sum* is updated by adding the loop variable *i* to it. Finally, the program prints the total sum using `System.out.println()`, displaying the result in the console.

SET-02(Intermediate)

Problem No: 01

Problem Name: Store and print 5 numbers in an array.

Source Code:

```
package print_array.c251222;

import java.util.Scanner;

public class Print_ArrayC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        for (int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

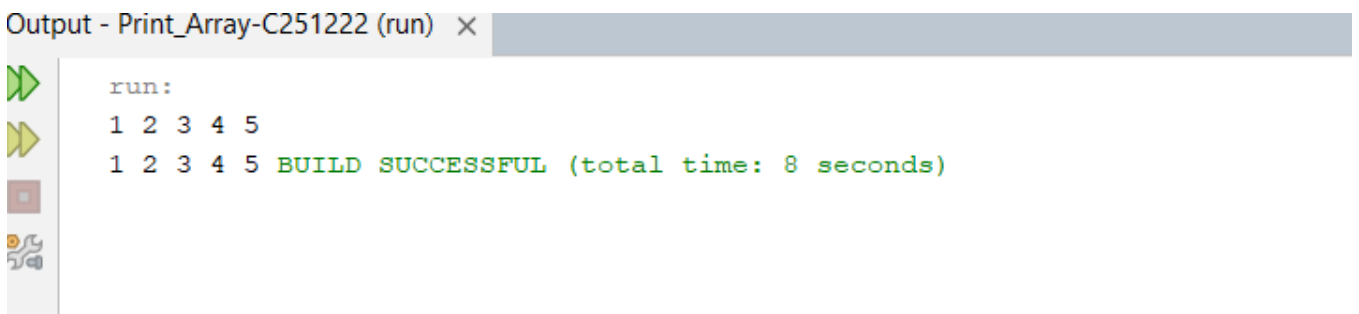
        for (int num : arr)

            System.out.print(num + " ");

    }

}
```

Output:



```
Output - Print_Array-C251222 (run) X
run:
1 2 3 4 5
1 2 3 4 5 BUILD SUCCESSFUL (total time: 8 seconds)
```

Approach:

In this program, the elements of an array are taken as input and then printed. First, it creates an integer array of size 5 and takes input values from the user using the Scanner class. Then, a for loop is used to store the input values into the array.

After that, a for-each loop is used to access each element of the array and print it using System.out.print(). Finally, the program displays all the array elements in a single line on the console.

Problem No: 02

Problem Name: Find the largest number in an array.

Source Code:

```
package largest_in_array.c251222;

import java.util.Scanner;

public class Largest_in_arrayC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++)

            arr[i] = sc.nextInt();

        int max = arr[0];

        for (int i = 1; i < n; i++) {

            if (arr[i] > max)

                max = arr[i];

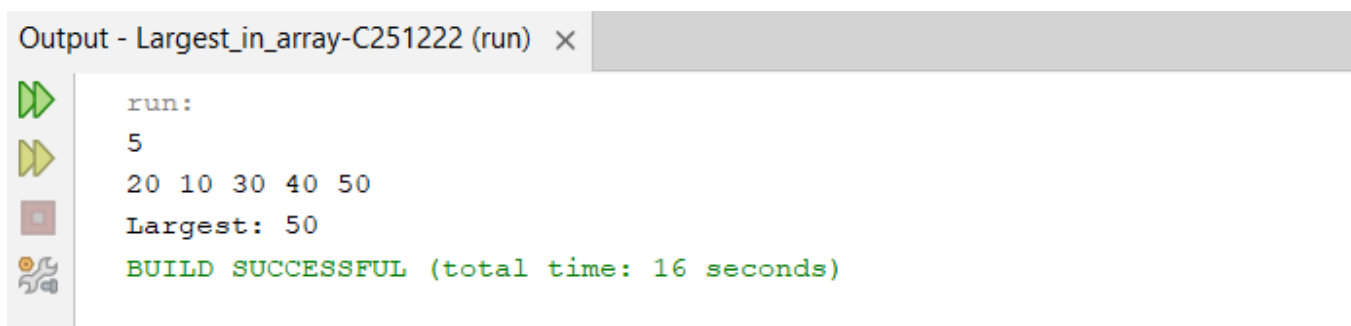
        }

        System.out.println("Largest: " + max);

    }

}
```

Output:



```
Output - Largest_in_array-C251222 (run) ×
run:
5
20 10 30 40 50
Largest: 50
BUILD SUCCESSFUL (total time: 16 seconds)
```

Approach:

This program finds the largest element in an array. First, it takes the size n and then reads n elements from the user using the Scanner class. The first element is assumed as the maximum. Then, a loop compares each element with the current maximum and updates it if a larger value is found. Finally, the program prints the largest element using `System.out.println()`.

Problem No: 03

Problem Name: Find the smallest number in an array.

Source Code:

```
package smallest_in_array.c251222;

import java.util.Scanner;

public class Smallest_in_arrayC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++)

            arr[i] = sc.nextInt();

        int min = arr[0];

        for (int i = 1; i < n; i++) {

            if (arr[i] < min)

                min = arr[i];

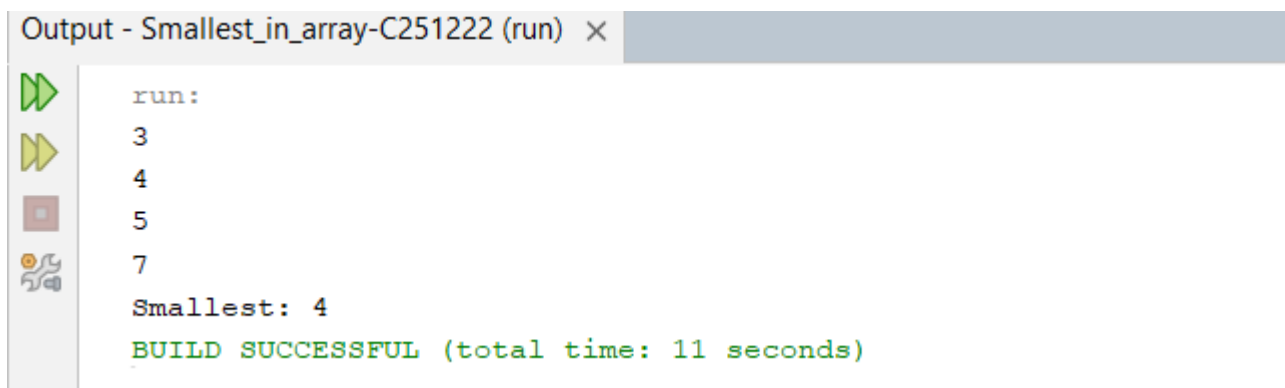
        }

        System.out.println("Smallest: " + min);

    }

}
```

Output:



```
Output - Smallest_in_array-C251222 (run) ×
run:
3
4
5
7
Smallest: 4
BUILD SUCCESSFUL (total time: 11 seconds)
```

Approach:

This program finds the smallest element in an array. It takes the size n and inputs n elements using the Scanner class, assuming the first element as the minimum. Then, it compares each element and updates the minimum if a smaller value is found. Finally, it prints the smallest value using `System.out.println()`.

Problem No: 04

Problem Name: Calculate the average of an array.

Source Code:

```
package avarrage.c251222;

import java.util.Scanner;

public class AvarrageC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        int sum = 0;

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

            sum += arr[i];

        }

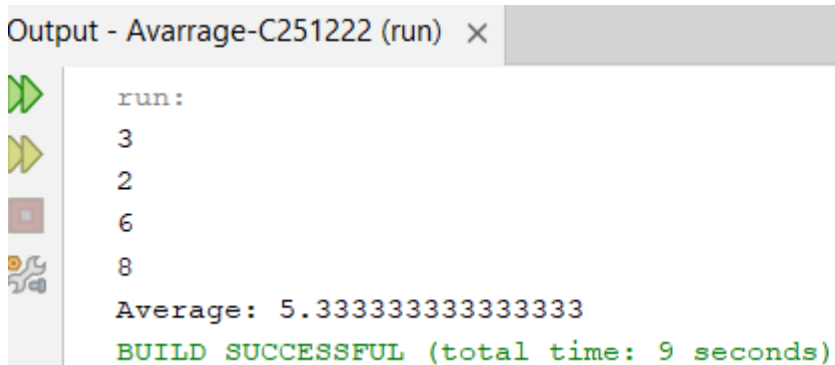
        double avg = (double) sum / n;

        System.out.println("Average: " + avg);

    }

}
```

Output:



```
Output - Avarrage-C251222 (run) ×
run:
3
2
6
8
Average: 5.333333333333333
BUILD SUCCESSFUL (total time: 9 seconds)
```

Approach:

This program is used to calculate the average of elements in an array. First, it takes an integer input *n* to determine the size of the array and then reads *n* values from the user using the Scanner class. While taking the input, it also calculates the total sum of all elements. After that, the average is calculated by dividing the sum by *n*, where type casting is used to get a decimal result. Finally, the program prints the average using `System.out.println()`, displaying the result in the console.

Problem No: 05

Problem Name: Count even and odd numbers in an array.

Source Code:

```
package count.c251222;

import java.util.Scanner;

public class CountC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        int even = 0, odd = 0;

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

            if (arr[i] % 2 == 0)

                even++;

            else

                odd++;

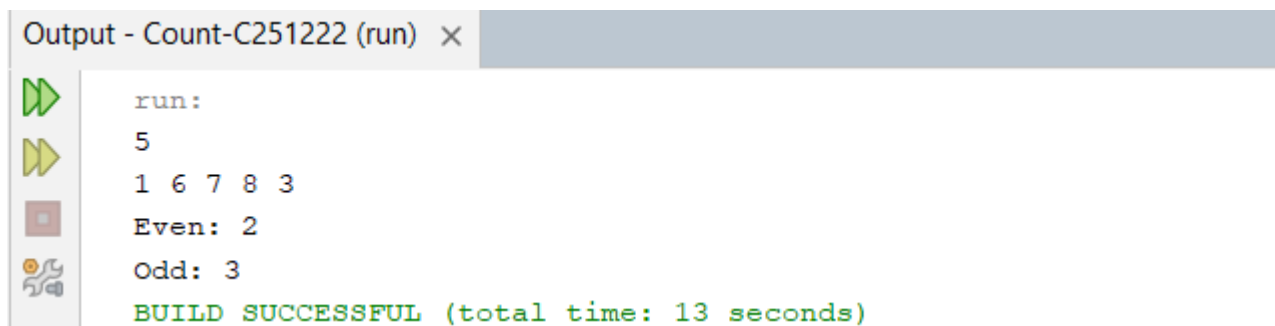
        }

        System.out.println("Even: " + even);

        System.out.println("Odd: " + odd);

    } }
```

Output:



```
Output - Count-C251222 (run) ×
run:
5
1 6 7 8 3
Even: 2
Odd: 3
BUILD SUCCESSFUL (total time: 13 seconds)
```

Approach:

This program counts the number of even and odd elements in an array. It takes the size n and reads n values from the user using the Scanner class. While taking input, each number is checked—if it is divisible by 2, the even counter is increased; otherwise, the odd counter is increased. Finally, the program prints the total even and odd counts.

Problem No: 06

Problem Name: Search for an element in an array.

Source Code:

```
package search.c251222;

import java.util.Scanner;

public class SearchC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++)

            arr[i] = sc.nextInt();

        int key = sc.nextInt();

        for (int i = 0; i < n; i++) {

            if (arr[i] == key) {

                System.out.println("Found at index " + i);

                return;

            }

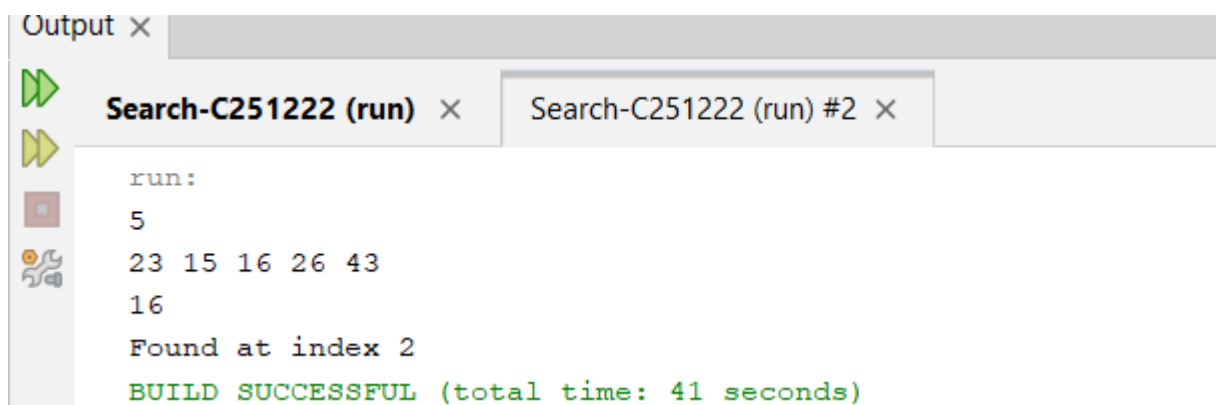
        }

        System.out.println("Not Found");

    }

}
```

Output:



Approach:

This program searches for a specific element in an array. First, it takes the size n and reads n elements using the Scanner class. Then it takes a key value to search in the array. After that, it checks each element one by one. If the key is found, it prints the index and stops the program. If the key is not found in the array, it prints "Not Found".

Problem No: 07

Problem Name: Sort an array (Ascending Order).

Source Code:

```
package sort.c251222;

import java.util.Scanner;
import java.util.Arrays;

public class SortC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++)

            arr[i] = sc.nextInt();

        Arrays.sort(arr);

        System.out.println("Sorted Array:");

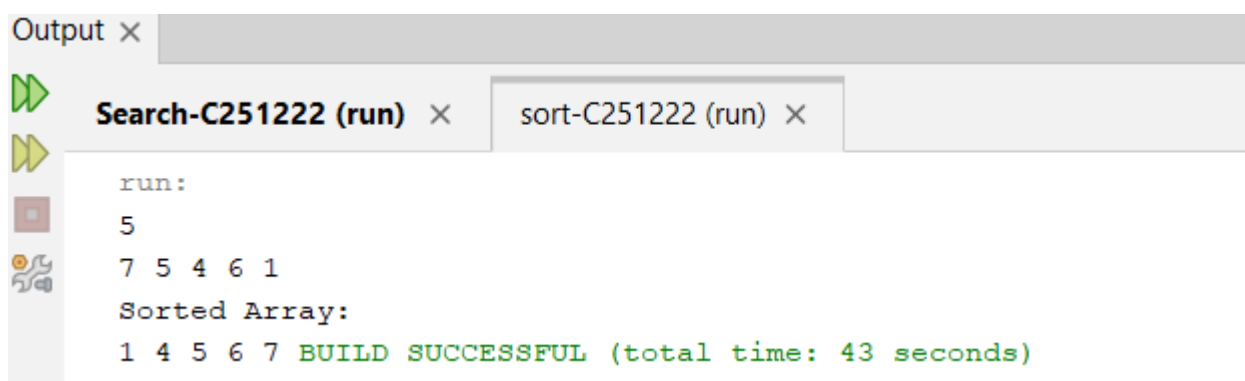
        for (int num : arr)

            System.out.print(num + " ");

    }

}
```

Output:



Approach:

This program sorts the elements of an array in ascending order. First, it takes the size *n* and reads *n* elements from the user using the Scanner class. Then, it uses the `Arrays.sort()` method to sort the array.

After sorting, the program prints the sorted array using a loop and displays the elements in order using `System.out.print()`.

Problem No: 08

Problem Name: Check if a string is a palindrome.

Source Code:

```
package palindrome.c251222;

import java.util.Scanner;

public class PalindromeC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {

            rev += str.charAt(i);

        }

        if (str.equals(rev))

            System.out.println("Palindrome");

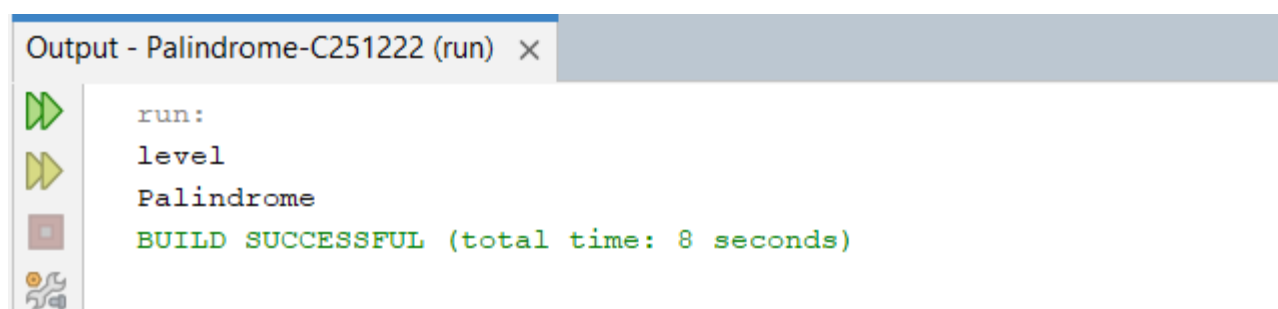
        else

            System.out.println("Not Palindrome");

    }

}
```

Output:



```
Output - Palindrome-C251222 (run) ×
run:
level
Palindrome
BUILD SUCCESSFUL (total time: 8 seconds)
```

Approach:

This program is used to check whether a given string is a palindrome or not. First, it takes a string input from the user using the Scanner class. Then, a new string rev is created, and the original string is reversed by adding characters from the end to the beginning using a loop.

After reversing, the program compares the original string with the reversed string using the equals() method. If both strings are equal, it prints “Palindrome”; otherwise, it prints “Not Palindrome”.

Problem No: 09

Problem Name: Count vowels and consonants in a string.

Source Code:

```
package vowels_consonants.c251222;

import java.util.Scanner;

public class Vowels_ConsonantsC251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine().toLowerCase();

        int vowels = 0, consonants = 0;

        for (int i = 0; i < str.length(); i++) {

            char ch = str.charAt(i);

            if (Character.isLetter(ch)) {

                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')

                    vowels++;

                else

                    consonants++;

            }

        }

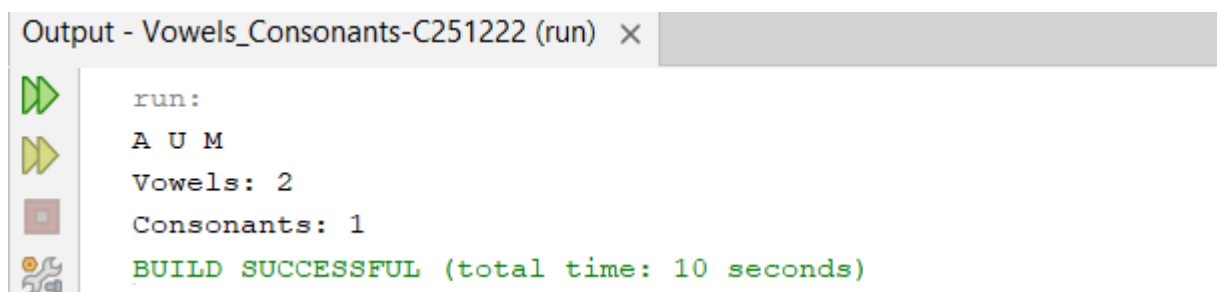
        System.out.println("Vowels: " + vowels);

        System.out.println("Consonants: " + consonants);

    }

}
```

Output:



```
Output - Vowels_Consonants-C251222 (run) x
run:
A U M
Vowels: 2
Consonants: 1
BUILD SUCCESSFUL (total time: 10 seconds)
```

Approach:

This program counts the number of vowels and consonants in a string. It takes a string input from the user and converts it to lowercase. Then, it checks each character: if it is a vowel (a, e, i, o, u), it increases the vowel count; otherwise, it increases the consonant count. Finally, it prints the total number of vowels and consonants.

Problem No: 10

Problem Name: Convert a string to uppercase and lowercase.

Source Code:

```
package uppercase_lowercase.c251222;

import java.util.Scanner;

public class Uppercase_LowercaseC251222 {

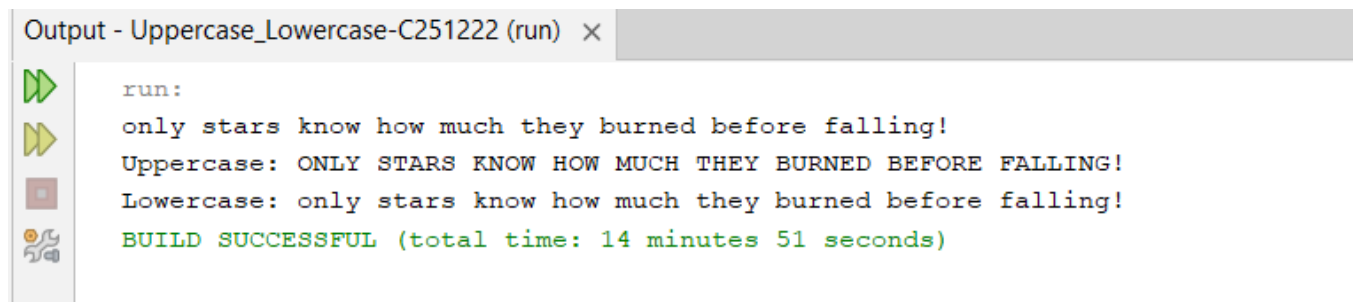
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        System.out.println("Uppercase: " + str.toUpperCase());
        System.out.println("Lowercase: " + str.toLowerCase());
    }
}
```

Output:

The screenshot shows a Java IDE's output window titled "Output - Uppercase_Lowercase-C251222 (run)". The output text is as follows:

```
run:
only stars know how much they burned before falling!
Uppercase: ONLY STARS KNOW HOW MUCH THEY BURNED BEFORE FALLING!
Lowercase: only stars know how much they burned before falling!
BUILD SUCCESSFUL (total time: 14 minutes 51 seconds)
```

Approach:

This program is used to convert a given string into uppercase and lowercase. First, it takes a string input from the user using the Scanner class. Then, it uses the toUpperCase() method to convert the string into uppercase letters and the toLowerCase() method to convert it into lowercase letters.

Finally, the program prints both the uppercase and lowercase versions of the string using System.out.println().

SET-02 (Descriptive Problems)

Problem No: 01

Source Code:

```
package set3_prob1.c251222;

import java.util.Scanner;

public class Set3_prob1C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int A, C;

        System.out.print("Enter A: ");

        A = sc.nextInt();

        System.out.print("Enter C: ");

        C = sc.nextInt();

        if ((A + C) % 2 == 0) {

            System.out.println("There exists B");

        } else {

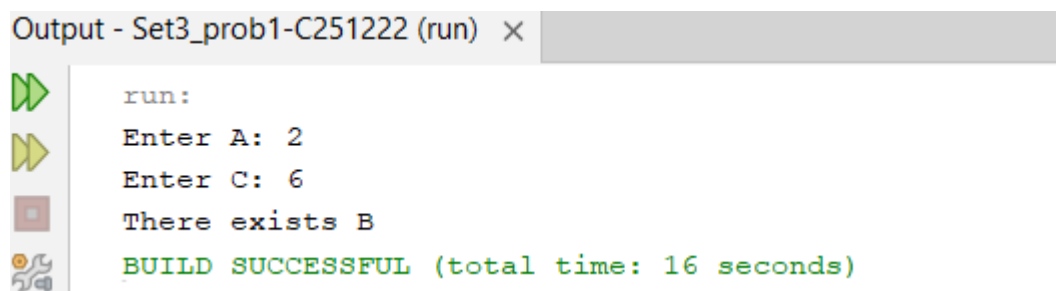
            System.out.println("There does not exist B");

        }

    }

}
```

Output:



```
Output - Set3_prob1-C251222 (run) ×
run:
Enter A: 2
Enter C: 6
There exists B
BUILD SUCCESSFUL (total time: 16 seconds)
```

Approach:

This program checks whether a value B can exist based on two given integers A and C. First, it takes two integer inputs A and C from the user using the Scanner class.

Then, it adds A and C and checks if the sum is even by using the condition $(A + C) \% 2 == 0$. If the condition is true, it prints that a value of B exists; otherwise, it prints that B does not exist.

Problem No: 02

Source Code:

```
package set3_prob2.c251222;

import java.util.Scanner;

public class Set3_prob2C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String s;

        System.out.print("Enter string: ");

        s = sc.nextLine();

        int size = s.length();

        int lastIndex = size - 1;

        if (size > 10) {

            System.out.print(s.charAt(0));

            System.out.print(size - 2);

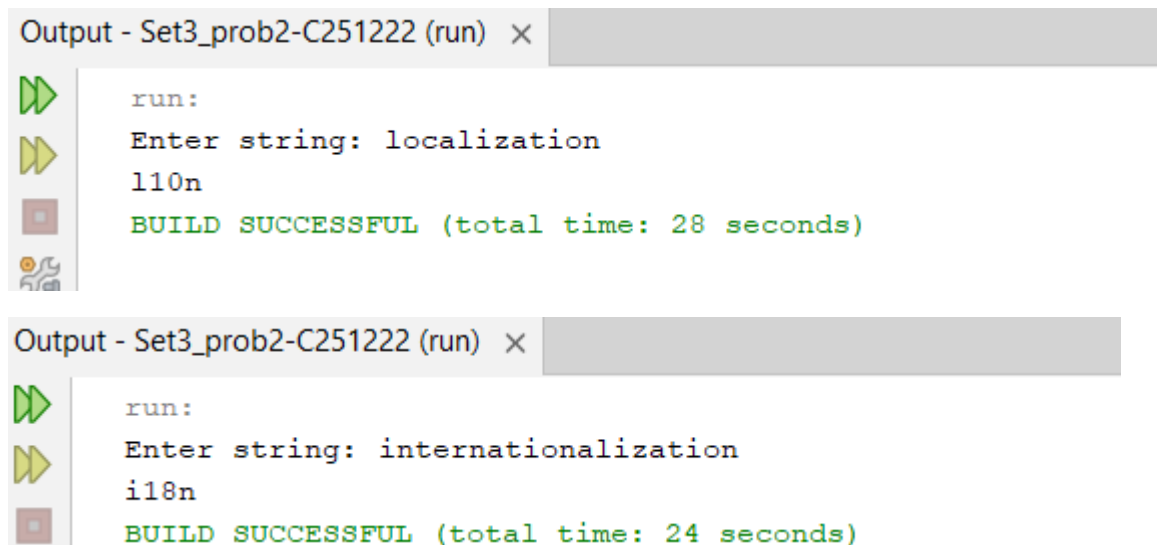
            System.out.println(s.charAt(lastIndex));

        } else {

            System.out.println(s);

        } } }
```

Output:



The image shows two screenshots of an IDE's output window. The first screenshot shows the program running with the input 'localization', resulting in the output 'l10n'. The second screenshot shows the program running with the input 'internationalization', resulting in the output 'i18n'. Both runs were successful and took approximately 28 and 24 seconds respectively.

```
Output - Set3_prob2-C251222 (run) x
run:
Enter string: localization
l10n
BUILD SUCCESSFUL (total time: 28 seconds)

Output - Set3_prob2-C251222 (run) x
run:
Enter string: internationalization
i18n
BUILD SUCCESSFUL (total time: 24 seconds)
```

Approach:

This program takes a string input and checks its length. If the length is greater than 10, it prints a shortened version by showing the first character, the number of characters in between, and the last character. Otherwise, it prints the original string.

Problem No: 03

Source Code:

```
package set03_prob03.c251222;

import java.util.Scanner;

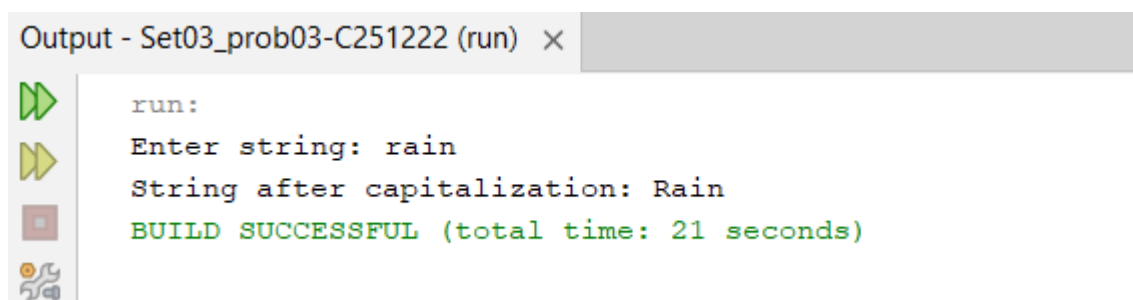
public class Set03_prob03C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s;
        System.out.print("Enter string: ");

        s = sc.nextLine();
        String result = s.substring(0, 1).toUpperCase() + s.substring(1);

        System.out.println("String after capitalization: " + result);
    }
}
```

Output:



```
Output - Set03_prob03-C251222 (run) ×
run:
Enter string: rain
String after capitalization: Rain
BUILD SUCCESSFUL (total time: 21 seconds)
```

Approach:

This program is used to capitalize the first letter of a given string. It first takes a string input from the user using the Scanner class. Then, it separates the first character and converts it to uppercase using `substring()` and `toUpperCase()`, while keeping the remaining part of the string unchanged.

After combining both parts, the program prints the modified string with the first letter capitalized using `System.out.println()`.

Problem No: 04

Source Code:

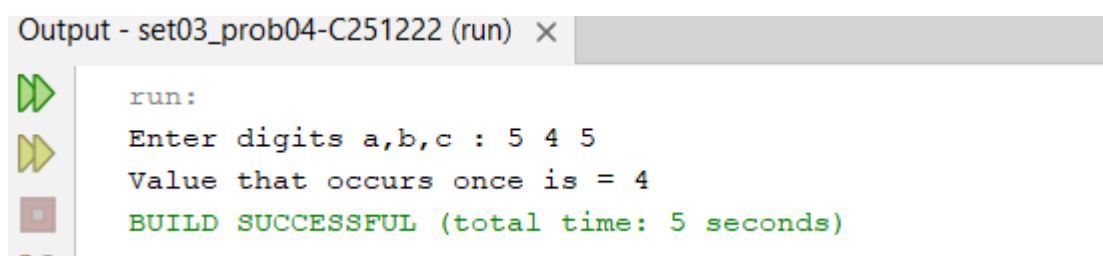
```
package set03_prob04.c251222;

import java.util.Scanner;

public class Set03_prob04C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a, b, c;
        System.out.print("Enter digits a,b,c : ");
        a = sc.nextInt();
        b = sc.nextInt();
        c = sc.nextInt();
        if (a == b) {
            System.out.println("Value that occurs once is = " + c);
        } else if (b == c) {
            System.out.println("Value that occurs once is = " + a);
        } else if (c == a) {
            System.out.println("Value that occurs once is = " + b);
        }
    }
}
```

Output:



```
Output - set03_prob04-C251222 (run) ×
run:
Enter digits a,b,c : 5 4 5
Value that occurs once is = 4
BUILD SUCCESSFUL (total time: 5 seconds)
```

Approach:

This program finds the value that occurs only once among three given numbers. First, it takes three integer inputs a, b, and c from the user using the Scanner class. Then, it compares the values to check which two numbers are equal. Based on the condition, it identifies and prints the number that is different (the one that occurs only once). The result is displayed using System.out.println().

Problem No: 05

Source Code:

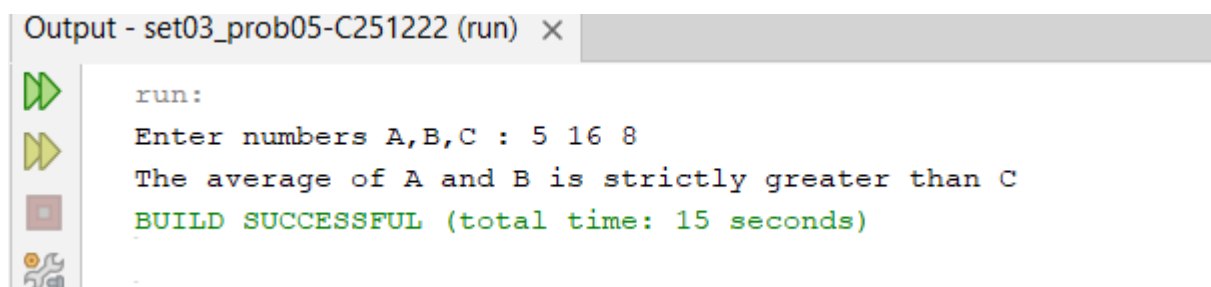
```
package set03_prob05.c251222;

import java.util.Scanner;

public class Set03_prob05C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a, b, c;
        System.out.print("Enter numbers A,B,C : ");
        a = sc.nextInt();
        b = sc.nextInt();
        c = sc.nextInt();
        if (((double) (a + b) / 2 > c) {
            System.out.println("The average of A and B is strictly greater than C");
        } else {
            System.out.println("The average of A and B is strictly not greater than C");
        }
    }
}
```

Output:



```
Output - set03_prob05-C251222 (run) ×
run:
Enter numbers A,B,C : 5 16 8
The average of A and B is strictly greater than C
BUILD SUCCESSFUL (total time: 15 seconds)
```

Approach:

This program compares the average of two numbers with a third number. First, it takes three integer inputs a, b, and c from the user using the Scanner class.

Then, it calculates the average of a and b and checks whether it is strictly greater than c. If the condition is true, it prints that the average is greater than c; otherwise, it prints that it is not greater.

Problem No: 06

Source Code:

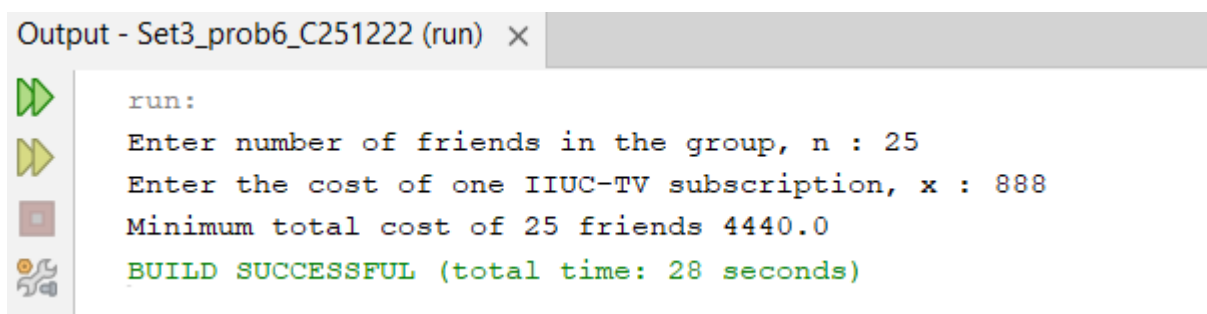
```
package set3_prob6_c251222;

import java.util.Scanner;

public class Set3_prob6_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n, x;
        System.out.print("Enter number of friends in the group, n : ");
        n = sc.nextInt();
        System.out.print("Enter the cost of one IIUC-TV subscription, x : ");
        x = sc.nextInt();
        double subscriptionNo;
        double total;
        subscriptionNo = (double) n / 6;
        total = Math.ceil(subscriptionNo) * x;
        System.out.println("Minimum total cost of " + n + " friends " + total);
    }
}
```

Output:



```
Output - Set3_prob6_C251222 (run) ×
run:
Enter number of friends in the group, n : 25
Enter the cost of one IIUC-TV subscription, x : 888
Minimum total cost of 25 friends 4440.0
BUILD SUCCESSFUL (total time: 28 seconds)
```

Approach:

This program calculates the minimum total cost for a group of friends to share a subscription. First, it takes two inputs from the user: the number of friends n and the cost of one subscription x . Then, it divides the number of friends by 6 to determine how many subscriptions are needed, and uses `Math.ceil()` to round up to the next whole number. Finally, it multiplies the required number of subscriptions by the cost and prints the minimum total cost using `System.out.println()`.

Problem No: 07

Source Code:

```
package set3_prob7_c251222;

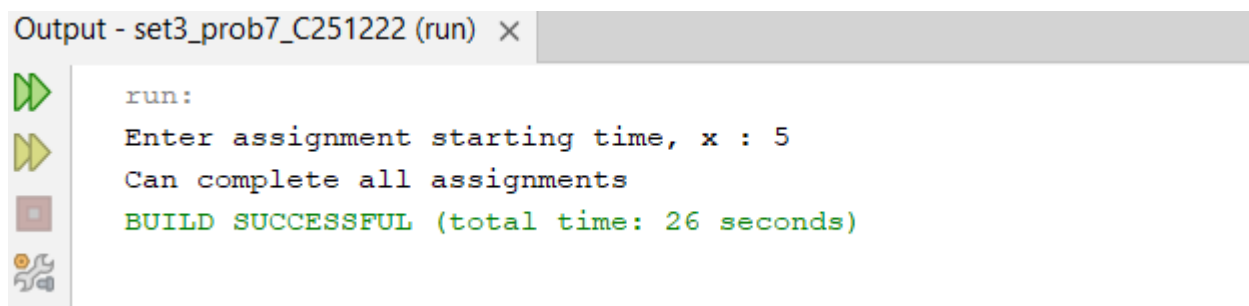
import java.util.Scanner;

public class Set3_prob7_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int x;
        System.out.print("Enter assignment starting time, x : ");
        x = sc.nextInt();
        if (x + 3 <= 10) {
            System.out.println("Can complete all assignments");
        } else {

            System.out.println("Can not complete all assignments");
        }
    }
}
```

Output:



```
Output - set3_prob7_C251222 (run) ×
run:
Enter assignment starting time, x : 5
Can complete all assignments
BUILD SUCCESSFUL (total time: 26 seconds)
```

Approach:

This program checks whether all assignments can be completed within a given time. First, it takes an integer input x , which represents the starting time, using the Scanner class.

Then, it adds 3 hours to the starting time and checks if the total time is less than or equal to 10. If the condition is true, it prints that all assignments can be completed; otherwise, it prints that they cannot be completed.

Problem No: 08

Source Code:

```
package set3_prob8_c251222;

import java.util.Scanner;

public class Set3_prob8_C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int x, y, z;

        System.out.print("Enter number of sections, X: ");

        x = sc.nextInt();

        System.out.print("Enter students per section, Y: ");

        y = sc.nextInt();

        System.out.print("Enter number of students passed, Z: ");

        z = sc.nextInt();

        double percent = ((double) z / (x * y)) * 100;

        System.out.println("Percentage of passing students = " + percent);

        if (percent > 50) {

            System.out.println("Percentage is strictly greater than 50%");

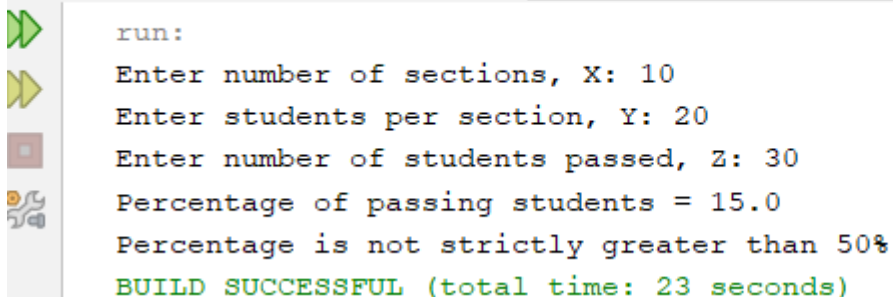
        } else {

            System.out.println("Percentage is not strictly greater than 50%");

        } } }
```

Output:

Output - set3_prob8_C251222 (run) ×



```
run:
Enter number of sections, X: 10
Enter students per section, Y: 20
Enter number of students passed, Z: 30
Percentage of passing students = 15.0
Percentage is not strictly greater than 50%
BUILD SUCCESSFUL (total time: 23 seconds)
```

Approach:

This program calculates the percentage of students who passed. It takes inputs for sections x , students per section y , and passed students z using the Scanner class. Then, it finds the total students and computes the percentage using $(z / (x * y)) * 100$. Finally, it prints the percentage and checks whether it is greater than 50%, displaying the corresponding message.

Problem No: 09

Source Code:

```
package set3_prob9_c251222;

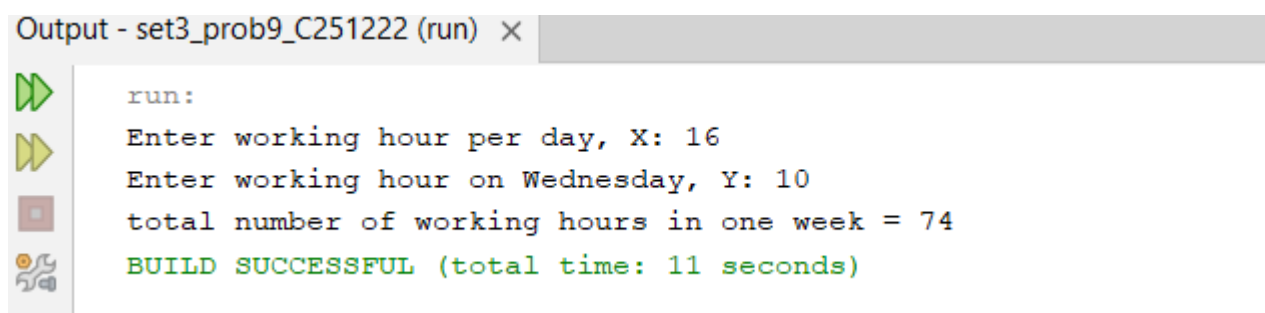
import java.util.Scanner;

public class Set3_prob9_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int x, y;
        System.out.print("Enter working hour per day, X: ");
        x = sc.nextInt();
        System.out.print("Enter working hour on Wednesday, Y: ");
        y = sc.nextInt();
        int total = (x * 4) + y;

        System.out.println("total number of working hours in one week = " + total);
    }
}
```

Output:



```
Output - set3_prob9_C251222 (run) ×
run:
Enter working hour per day, X: 16
Enter working hour on Wednesday, Y: 10
total number of working hours in one week = 74
BUILD SUCCESSFUL (total time: 11 seconds)
```

Approach:

In this program, the total working hours for a week are determined using simple arithmetic. It first takes two inputs from the user: working hours per day x and working hours on Wednesday y using the Scanner class.

Then, it calculates the total hours using the formula $(x * 4) + y$, assuming four regular working days and a different value for Wednesday. Finally, it displays the total working hours using `System.out.println()`.

Problem No: 10

Source Code:

```
package set3_prob10.c251222;

import java.util.Scanner;

public class Set3_prob10C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int B1, B2, B3;

        System.out.print("Enter B1: ");

        B1 = sc.nextInt();

        System.out.print("Enter B2: ");

        B2 = sc.nextInt();

        System.out.print("Enter B3: ");

        B3 = sc.nextInt();

        if ((B1 == 0 && B2 == 0) || (B2 == 0 && B3 == 0) || (B3 == 0 && B1 == 0)) {

            System.out.println("Water filling time");

        } else {

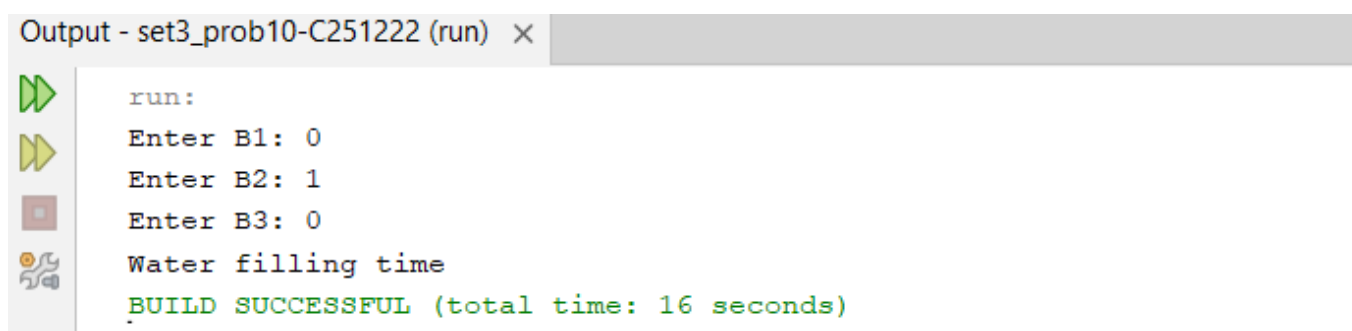
            System.out.println("Not now");

        }

    }

}
```

Output:



```
Output - set3_prob10-C251222 (run) ×
run:
Enter B1: 0
Enter B2: 1
Enter B3: 0
Water filling time
BUILD SUCCESSFUL (total time: 16 seconds)
```

Approach:

This program checks whether the condition for water filling is satisfied. It takes three integer inputs B1, B2, and B3 from the user using the Scanner class.

Then, it checks if at least two of the values are zero using logical conditions. If the condition is true, it prints “Water filling time”; otherwise, it prints “Not now”.

Problem No: 11

Source Code:

```
package set3_prob11_c251222;

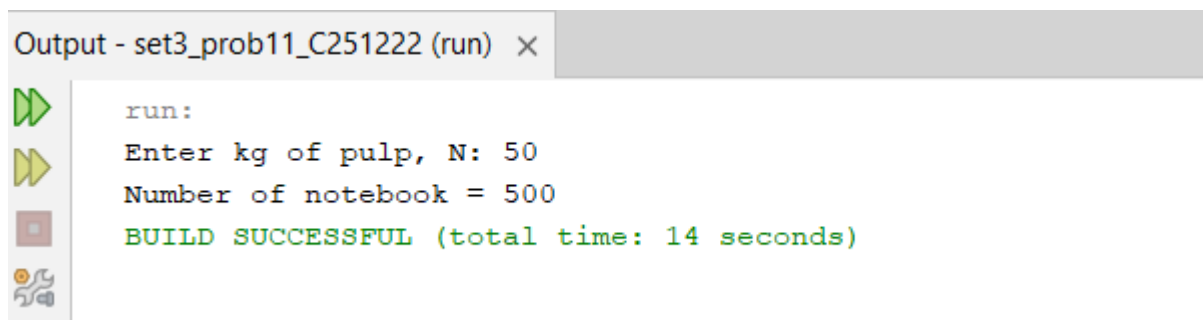
import java.util.Scanner;

public class Set3_prob11_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double N;
        System.out.print("Enter kg of pulp, N: ");
        N = sc.nextDouble();
        double total = N * 10;

        System.out.println("Number of notebook = "+(int)Math.floor(total));
    }
}
```

Output:



```
Output - set3_prob11_C251222 (run) ×
run:
Enter kg of pulp, N: 50
Number of notebook = 500
BUILD SUCCESSFUL (total time: 14 seconds)
```

Approach:

This program calculates the number of notebooks that can be produced from a given amount of pulp. First, it takes a double input N from the user using the Scanner class, where N represents the amount of pulp in kilograms. Then, the program uses a simple formula to estimate the total number of notebooks that can be produced by multiplying N by 10. Since the number of notebooks must be a whole number, the program applies Math.floor() to round down the result to the nearest integer.

Finally, it prints the total number of notebooks using System.out.println(), displaying the final result in the console. This approach ensures that the program only considers complete notebooks and ignores any fractional part of the calculation.

Problem No: 12

Source Code:

```
package set3_prob12_c251222;

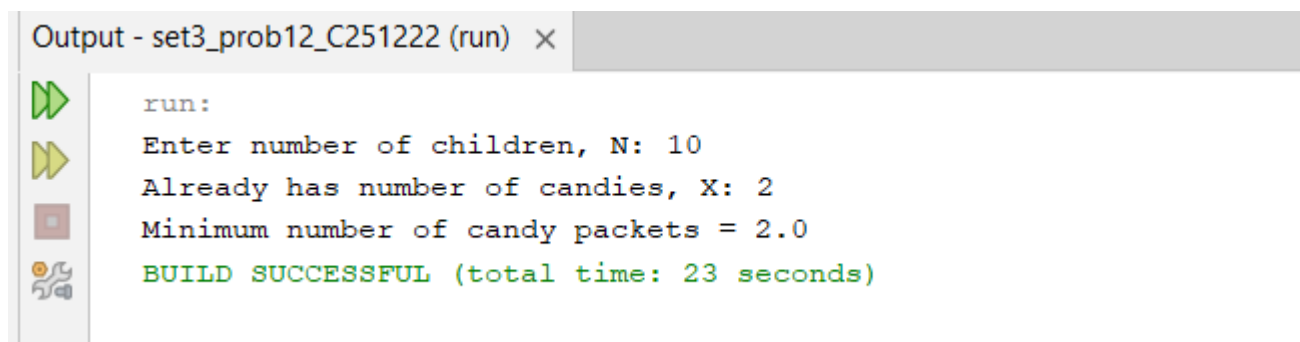
import java.util.Scanner;

public class Set3_prob12_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N, X;
        System.out.print("Enter number of children, N: ");
        N = sc.nextInt();
        System.out.print("Already has number of candies, X: ");
        X = sc.nextInt();
        double candyPacketNo = (double) (N - X) / 4;

        System.out.println("Minimum number of candy packets = " + Math.ceil(candyPacketNo));
    }
}
```

Output:



```
Output - set3_prob12_C251222 (run) ×
run:
Enter number of children, N: 10
Already has number of candies, X: 2
Minimum number of candy packets = 2.0
BUILD SUCCESSFUL (total time: 23 seconds)
```

Approach:

This program is used to find the minimum number of candy packets required to distribute candies among children. It takes two integer inputs from the user using the Scanner class: the total number of children N and the number of candies already available X. Then, it calculates the remaining candies needed by subtracting X from N, and divides the result by 4 since each packet contains 4 candies. To ensure that all children receive enough candies, the program uses Math.ceil() to round up the result to the next whole number. Finally, it prints the minimum number of candy packets required using System.out.println(), ensuring proper distribution of candies.

Problem No: 13

Source Code:

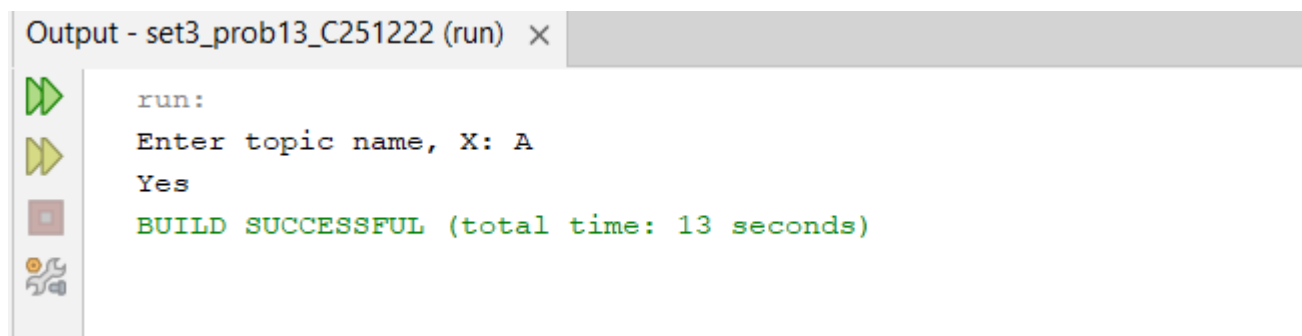
```
package set3_prob13_c251222;

import java.util.Scanner;

public class Set3_prob13_C251222 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        char x;
        System.out.print("Enter topic name, X: ");
        x = sc.next().charAt(0);
        if (x == 'A' || x == 'B' || x == 'C') {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output:



```
Output - set3_prob13_C251222 (run) ×
run:
Enter topic name, X: A
Yes
BUILD SUCCESSFUL (total time: 13 seconds)
```

Approach:

This program checks whether a given topic is valid based on a character input. It takes a character input X from the user using the Scanner class.

Then, it checks if the input character is either A, B, or C. If the condition is true, it prints “Yes”; otherwise, it prints “No”. Finally, the result is displayed using System.out.println().

Problem No: 14

Source Code:

```
package set3_prob14_c251222;

import java.util.Scanner;

public class Set3_prob14_C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int P, Q, R, S;

        System.out.print("Enter profit of company A, P: ");

        P = sc.nextInt();

        System.out.print("Enter profit of company B, Q: ");

        Q = sc.nextInt();

        System.out.print("Enter profit of company C, R: ");

        R = sc.nextInt();

        System.out.print("Enter profit of company D, S: ");

        S = sc.nextInt();

        if ((P > (Q + R + S)) || (Q > (R + S + P)) || (R > (S + P + Q)) || (S > (P + Q + R))) {

            System.out.println("There is a monopoly");

        } else {

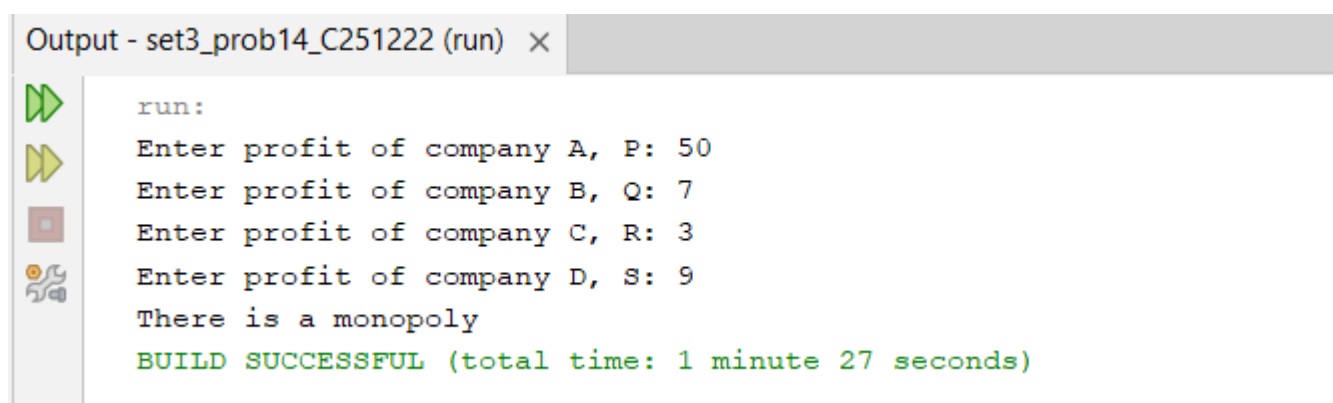
            System.out.println("There is no monopoly");

        }

    }

}
```

Output:



The screenshot shows a window titled "Output - set3_prob14_C251222 (run)". On the left is a vertical toolbar with icons for running, stepping through, and debugging. The main area displays the program's output: "run:", "Enter profit of company A, P: 50", "Enter profit of company B, Q: 7", "Enter profit of company C, R: 3", "Enter profit of company D, S: 9", "There is a monopoly", and "BUILD SUCCESSFUL (total time: 1 minute 27 seconds)".

Approach:

This program checks for a monopoly among four companies. It takes four profit values as input using the Scanner class. Then, it checks if any one company's profit is greater than the sum of the other three. If the condition is true, it prints "There is a monopoly"; otherwise, it prints "There is no monopoly".

Problem No: 15

Source Code:

```
package set3_prob15_c251222;

import java.util.Scanner;

public class Set3_prob15_C251222 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int A, B, C;

        System.out.print("Alice's required temperature A: ");

        A = sc.nextInt();

        System.out.print("Bob's required temperature B: ");

        B = sc.nextInt();

        System.out.print("Charlie's required temperature C: ");

        C = sc.nextInt();

        if (Math.max(A, C) <= B) {

            System.out.println("Yes");

        } else {

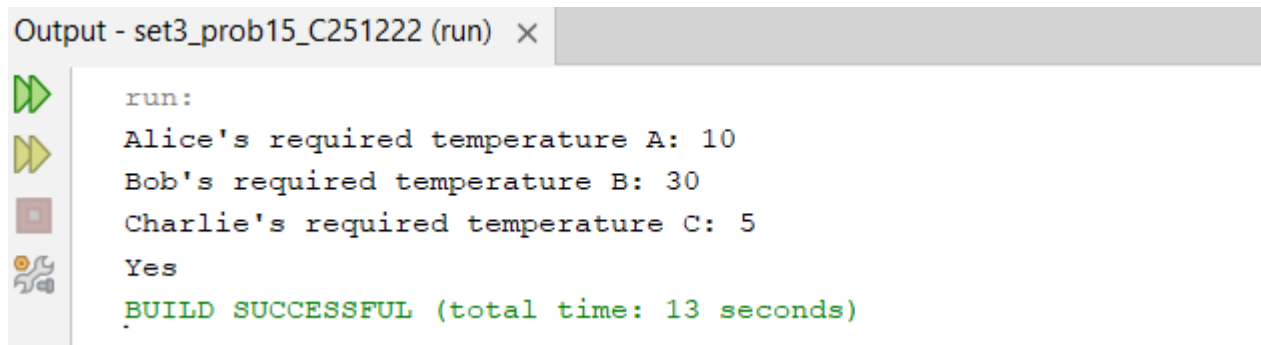
            System.out.println("No");

        }

    }

}
```

Output:



```
Output - set3_prob15_C251222 (run) ×
run:
Alice's required temperature A: 10
Bob's required temperature B: 30
Charlie's required temperature C: 5
Yes
BUILD SUCCESSFUL (total time: 13 seconds)
```

Approach:

This program checks whether Bob’s required temperature is suitable compared to Alice and Charlie. It takes three integer inputs A, B, and C using the Scanner class. Then, it finds the maximum of A and C and compares it with B. If B is greater than or equal to the maximum of A and C, it prints “Yes”; otherwise, it prints “No”.